

AI TRAVEL HUB

Product Backlog

16	8	75	6
User Stories	With Acceptance Criteria	Total Story Points	Epics

AUTH & ONBOARDING	TRIP MANAGEMENT	AI FEATURES	FINANCE	REAL-TIME	NOTIFICATIONS
-------------------	-----------------	-------------	---------	-----------	---------------

Backlog Overview — All Stories

ID	Title	Epic	Priority	SP	Has AC
US-001	Social Login with Google	AUTH & ONBOARDING	CRITICAL	5	YES
US-002	Create a Trip Group	TRIP MANAGEMENT	CRITICAL	3	YES
US-003	Join Group via Invite Code	TRIP MANAGEMENT	CRITICAL	2	YES
US-004	Create a Trip	TRIP MANAGEMENT	CRITICAL	3	-
US-005	AI-Generated Itinerary (Pathfinder Agent)	AI FEATURES	CRITICAL	13	YES
US-006	Edit and Reorder Itinerary Items	TRIP MANAGEMENT	HIGH	5	-
US-007	Real-Time Group Chat	REAL-TIME	CRITICAL	8	YES
US-008	Collaborative Trip Checklist	REAL-TIME	HIGH	5	YES
US-009	Log a Trip Expense	FINANCE	CRITICAL	5	YES
US-010	Split Expenses Equally or Custom	FINANCE	HIGH	5	-
US-011	View Debt Summary	FINANCE	HIGH	3	YES
US-012	Attach Receipt Photo to Expense	FINANCE	MEDIUM	3	-
US-013	AI Concierge — In-Trip Assistance	AI FEATURES	HIGH	8	YES
US-014	Push Notifications for Key Events	NOTIFICATIONS	HIGH	8	YES
US-015	Currency Conversion for International Groups	FINANCE	MEDIUM	3	-
US-016	Group Presence & Typing Indicators	REAL-TIME	LOW	3	-

Stories marked with 'YES' in the Has AC column include detailed acceptance criteria in their individual cards below. Story points follow the Fibonacci sequence: 1, 2, 3, 5, 8, 13. Stories above 8 SP should be broken down before sprint planning.

User Story Cards

US-001	Social Login with Google	AUTH & ONBOARDING	CRITICAL
Story Points	5 SP		
Persona	New user who wants to sign up quickly without creating another password		
User Story	As a new user, I want to sign in with my Google account so that I can join the platform in seconds without managing a separate password.		

Use django-allauth. On first login, create UserProfile with preferred_currency defaulting to EUR. JWT tokens: access 15 min, refresh 30 days.

Acceptance Criteria	
AC-1	Given the user taps 'Continue with Google', when Google OAuth completes successfully, then the user is redirected to the dashboard and a JWT access token + refresh token are issued.
AC-2	Given the OAuth callback succeeds, when the email does not exist in the database, then a new User and UserProfile are created automatically.
AC-3	Given the OAuth callback succeeds, when the email already exists, then the user is logged into the existing account without creating a duplicate.
AC-4	Given the OAuth flow fails or is cancelled, when the user returns to the app, then a clear error message is shown and no account is created.
AC-5	Given the access token has expired, when the client sends a refresh request with a valid refresh token, then a new access token is returned within 500ms.
AC-6	Given a refresh token older than 30 days, when the client attempts to refresh, then a 401 is returned and the user must log in again.

US-002	Create a Trip Group	TRIP MANAGEMENT	CRITICAL
Story Points	3 SP		
Persona	Trip organizer planning a group vacation		
User Story	As a trip organizer, I want to create a named group with a shareable invite code so that I can bring my friends into one shared space.		

Invite code: 8 alphanumeric characters, auto-generated, unique. Creator is automatically assigned the 'admin' role in GroupMembership.

Acceptance Criteria	
AC-1	Given the user submits a valid group name (3-100 chars), when the form is submitted, then a Group is created, a unique 8-character invite code is generated, and the creator is added as 'admin'.

AC-2	Given the group is created, when the user views the group settings, then the invite code and a 'Copy link' button are visible.
AC-3	Given the user submits a group name shorter than 3 characters or longer than 100, when the form is submitted, then a validation error is shown and no group is created.
AC-4	Given a group already exists with the same invite code, when the system generates a new code, then it retries until a unique code is produced.

US-003	Join Group via Invite Code	TRIP MANAGEMENT	CRITICAL
Story Points	2 SP		
Persona	Friend who received an invite link		
User Story	As an invited friend, I want to join a group by entering or clicking an invite code so that I can see the trip plans and collaborate immediately.		

Acceptance Criteria			
AC-1	Given a logged-in user enters a valid 8-character invite code, when they tap 'Join', then a GroupMembership record is created with role 'member' and the user is redirected to the group dashboard.		
AC-2	Given a user enters an invalid or expired invite code, when they tap 'Join', then a descriptive error message is shown and no membership is created.		
AC-3	Given a user who is already a member of the group enters its invite code, when they tap 'Join', then the system shows 'You are already in this group' and does not create a duplicate membership.		
AC-4	Given a new member joins via invite link, when the join completes, then all other online group members receive a real-time WebSocket notification: 'Alex joined the group'.		

US-004	Create a Trip	TRIP MANAGEMENT	CRITICAL
Story Points	3 SP		
Persona	Group admin setting up the trip details		
User Story	As a group admin, I want to create a trip with a destination, dates, and budget so that the group has a shared planning context.		

A group can have multiple trips (e.g. summer trip, winter trip). Trip dates drive the itinerary day numbering.

US-005	AI-Generated Itinerary (Pathfinder Agent)	AI FEATURES	CRITICAL
--------	---	-------------	----------

Story Points	13 SP
Persona	Group member who wants a ready-made day-by-day plan
User Story	As a group member, I want to ask the AI to generate a full itinerary for our trip so that I don't have to research and plan every activity manually.

Model: Gemini 2.0 Flash. Processing is async via Celery. AI response must stream token-by-token via WebSocket — user sees text appearing progressively. Rate limit: 3 AI requests/minute per user, 10/hour.

Acceptance Criteria	
AC-1	Given the user sends a planning request (e.g. '7 days in Tokyo, budget 1500 EUR'), when the request is submitted, then the server returns { task_id, status: 'queued' } within 300ms and the UI shows a 'Generating...' streaming indicator.
AC-2	Given the Celery task is processing, when Gemini starts returning tokens, then each token chunk is pushed via WebSocket to all online group members within 100ms of being received from the API.
AC-3	Given the AI finishes generating, when the task completes, then ItineraryItem records are created in the database (one per activity) with correct day, order, title, location, and start_time fields populated.
AC-4	Given the itinerary is saved, when any group member opens the trip view, then all itinerary items are displayed grouped by day in the correct order.
AC-5	Given the user exceeds 3 AI requests in 60 seconds, when they submit another request, then a 429 response is returned with a message showing when they can try again.
AC-6	Given the Gemini API call fails after 3 retries, when the task is moved to the Dead Letter Queue, then a WebSocket message is pushed to the user: 'AI could not complete the request. Please try again.'

US-006	Edit and Reorder Itinerary Items	TRIP MANAGEMENT	HIGH
Story Points	5 SP		
Persona	Group member who wants to customize the AI-generated plan		
User Story	As a group member, I want to edit, reorder, and delete itinerary items so that I can adjust the AI's suggestions to fit our preferences.		

Changes must be broadcast via WebSocket to all group members in real time. Use optimistic UI updates on the client side.

US-007	Real-Time Group Chat	REAL-TIME	CRITICAL
Story Points	8 SP		
Persona	Group member coordinating with friends		

User Story	As a group member, I want to send and read messages in a group chat so that we can coordinate plans without switching to another app.
------------	---

Django Channels with Upstash Redis as channel layer. Messages are persisted to PostgreSQL. Implement reconnect logic on the client (1s, 2s, 4s backoff).

Acceptance Criteria	
AC-1	Given a user types a message and presses send, when the WebSocket is connected, then the message appears in the chat for all online group members within 200ms.
AC-2	Given a user opens the chat, when the view loads, then the last 50 messages are loaded from the database in chronological order.
AC-3	Given a user's WebSocket connection drops, when the client detects disconnection, then it automatically attempts to reconnect at 1s, 2s, and 4s intervals and shows a 'Reconnecting...' indicator.
AC-4	Given a user is offline when a message is sent, when they reconnect and open the chat, then all messages sent during their absence are visible.
AC-5	Given the AI finishes generating an itinerary, when the result is ready, then an automated message appears in the chat: 'Pathfinder generated a new itinerary. Tap to view.'

US-008	Collaborative Trip Checklist	REAL-TIME	HIGH
Story Points	5 SP		
Persona	Group member tracking pre-trip tasks		
User Story	As a group member, I want to manage a shared checklist of tasks so that everyone can see what still needs to be done before the trip.		

When a member ticks a box, it turns green for everyone instantly via WebSocket. Store completed_at and completed_by for accountability.

Acceptance Criteria	
AC-1	Given a group admin adds a checklist item, when the item is saved, then it appears in the checklist for all group members within 300ms via WebSocket push.
AC-2	Given a user marks a checklist item as complete, when the update is saved, then the item visually shows as completed (green) for all online group members within 200ms, displaying the name of who completed it.
AC-3	Given multiple users attempt to complete the same item simultaneously, when the server processes both requests, then only one completion is recorded and no duplicate entries are created.
AC-4	Given a user opens the checklist while offline, when they reconnect, then the checklist reflects the latest state from the server.

US-009	Log a Trip Expense	FINANCE	CRITICAL
--------	--------------------	---------	----------

Story Points	5 SP
Persona	Group member who paid for a shared activity
User Story	As a group member, I want to log an expense I paid for so that the group can track shared costs and settle debts at the end of the trip.

Amount stored as DecimalField(max_digits=10, decimal_places=2) — never FloatField. Currency is ISO 4217 (EUR, USD, RON, etc.).

Acceptance Criteria	
AC-1	Given the user fills in a valid amount, currency, and description, when they submit the expense form, then a Transaction record is created and a DebtSplit record is created for each group member with status 'pending'.
AC-2	Given the user enters an amount with more than 2 decimal places, when the form is submitted, then the amount is rounded to 2 decimal places before saving.
AC-3	Given a transaction is saved, when the group's expense list is viewed, then the new transaction appears at the top, showing payer name, amount, currency, and description.
AC-4	Given a user enters a negative or zero amount, when the form is submitted, then a validation error is shown and no transaction is created.
AC-5	Given the transaction is logged, when the debt summary is viewed, then each group member's running balance is updated to reflect the new split.

US-010	Split Expenses Equally or Custom	FINANCE	HIGH
Story Points	5 SP		
Persona	Group member calculating who owes what		
User Story	As a group member, I want to split an expense equally or assign custom amounts per person so that debt tracking is accurate even when someone skipped an activity.		

Rounding: when splitting equally and the amount is not divisible evenly, the remainder cent is assigned to the payer. This prevents systematic rounding errors over many transactions.

US-011	View Debt Summary	FINANCE	HIGH
Story Points	3 SP		
Persona	Group member wanting to know who owes what		
User Story	As a group member, I want to see a clear summary of who owes money to whom so that settling up at the end of the trip is fast and unambiguous.		

Use debt simplification algorithm: instead of showing 10 raw transactions, reduce to the minimum number of transfers needed. E.g. if A owes B 20 and B owes C 20, show A owes C 20 directly.

Acceptance Criteria	
AC-1	Given the trip has multiple transactions, when a user opens the Debt Summary screen, then they see a simplified list of net transfers (minimized number of payments) rather than a raw transaction log.
AC-2	Given a user has no outstanding debts, when they open the summary, then their balance shows 0.00 and a 'All settled up!' indicator is shown.
AC-3	Given transactions exist in multiple currencies, when the debt summary is rendered, then all amounts are shown in each member's preferred currency using the latest exchange rate.
AC-4	Given a DebtSplit is marked as 'paid', when the summary is refreshed, then that debt no longer appears in the outstanding balance.

US-01 2	Attach Receipt Photo to Expense	FINANCE	MEDIUM
Story Points	3 SP		
Persona	Group member who wants proof of a shared expense		
User Story	As a group member, I want to attach a photo of a receipt to an expense so that there is a verifiable record and no disputes about the amount.		

Direct browser-to-R2 upload via presigned URL (never proxied through Django). Max file size: 5MB. Accepted formats: JPEG, PNG, WebP. receipt_url is stored on the Transaction model.

US-01 3	AI Concierge — In-Trip Assistance	AI FEATURES	HIGH
Story Points	8 SP		
Persona	Group member needing local help during the trip		
User Story	As a group member during the trip, I want to ask the AI for local recommendations, emergency contacts, or travel tips so that I can resolve issues quickly without leaving the app.		

Concierge Agent context window: current trip location, last 5 messages (~500 tokens). Inject timezone and local date/time so answers are contextually relevant.

Acceptance Criteria	
AC-1	Given a user asks a local question (e.g. 'nearest pharmacy open now'), when the Concierge Agent processes the request, then a relevant answer is streamed back within 3 seconds for 95% of requests.

AC-2	Given a question is outside the travel domain, when the Dispatcher classifies the intent, then it responds with a polite redirect: 'I am specialized in travel assistance. How can I help with your trip?'
AC-3	Given the Concierge response includes an itinerary suggestion, when the user taps 'Add to itinerary', then the suggested activity is appended as an ItineraryItem and visible to the whole group.

US-01 4	Push Notifications for Key Events	NOTIFICATIONS	HIGH
Story Points	8 SP		
Persona	Group member who has the app in the background		
User Story	As a group member, I want to receive push notifications for important events so that I stay informed even when the app is not open.		

FCM for Android and iOS. Web Push API for desktop browsers. Push token stored in UserProfile.push_token. Users can configure notification preferences per event type.

Acceptance Criteria	
AC-1	Given an AI itinerary finishes generating, when the requesting user has a valid push token, then a push notification is sent within 5 seconds of task completion with the message 'Your itinerary for [destination] is ready!'
AC-2	Given a new expense is logged by another group member, when the affected debtors have notifications enabled, then each debtor receives a push notification showing the payer, amount, and description.
AC-3	Given a user has disabled expense notifications in their settings, when a new expense is logged, then no push notification is sent to that user but the in-app badge is still updated.
AC-4	Given a push token is stale or invalid, when the FCM send fails with a 'token not registered' error, then the push_token field is cleared from UserProfile and no further send attempts are made.

US-01 5	Currency Conversion for International Groups	FINANCE	MEDIUM
Story Points	3 SP		
Persona	Group member from a different country using a different currency		
User Story	As a group member, I want expenses logged in foreign currencies to be automatically converted to my preferred currency so that I always understand the real cost.		

Use Frankfurter API (frankfurter.app) — free, no API key required, updated daily. Cache exchange rates for 6 hours in Redis to avoid rate limits. Never convert the stored amount — only convert for display.

US-016	Group Presence & Typing Indicators	REAL-TIME	LOW
Story Points	3 SP		
Persona	Group member chatting and collaborating in real time		
User Story	As a group member, I want to see who is currently online and when someone is typing so that the chat feels like a real conversation.		

Presence state managed via Django Channels groups. Typing indicators: debounced at 300ms, cleared after 3 seconds of inactivity. Do not persist presence state to the database.